SYSTEM FOR SECURE COMMUNICATION USING COVERT SIDE-CHANNELS EMBEDDED IN A HIGH-CARDINALITY PARALLEL DATA STREAM

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Title: SYSTEM FOR SECURE COMMUNICATION USING COVERT SIDE-CHANNELS EMBEDDED IN A HIGH-CARDINALITY PARALLEL DATA STREAM

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "Be Water, Be Light," the contents of which are hereby incorporated by reference in their entirety. The system described herein is an improvement upon the high-cardinality parallel data bus described therein.

Abstract

A system and method for enhancing the security of a high-cardinality, parallel data communication channel by embedding one or more covert side-channels within the primary data

stream. These side-channels exist on a separate logical layer but share the same physical medium as the primary data. The side-channels are used to securely transmit sensitive information, such as cryptographic keys for forward secrecy or strategic portions of a larger message. Security is further enhanced by a method of phase-shifting the transmission of the side-channel data relative to the primary data it pertains to, making interception and reconstruction by an attacker computationally infeasible without knowledge of the phase-shifting parameters.

Background of the Invention

High-speed data communication relies on the secure transmission of both data and the cryptographic keys used to encrypt it. A common vulnerability in many communication protocols is that the data and the keys are transmitted through the same logical channel, or through separate channels that can be independently monitored. An attacker who can intercept both the encrypted data and the associated key can easily decrypt the message. Furthermore, traditional encryption schemes can be vulnerable if a single key is used to encrypt a large volume of data.

Summary of the Invention

The present invention provides a system for embedding covert, logically separate side-channels within a primary, high-cardinality parallel data stream. These side-channels utilize the same physical transmission medium as the primary data stream, but are not readily detectable by an observer of the primary data. In a preferred embodiment using a parallel optical interface, the primary data is encoded in the wavelengths of the light pixels, while the side-channel data is encoded in another property, such as the relative spatial positioning of those pixels.

These side-channels are used to transmit sensitive data, such as encryption keys or critical fragments of a larger message. By transmitting a portion of a message through the primary channel and a strategic portion through a covert side-channel, the system ensures that an attacker must successfully intercept and decode both channels to reconstruct the original message.

The system's security is further enhanced by a method of "phase-shifting" the side-channel data. The encryption key or data fragment corresponding to a primary data frame N is not

transmitted in the side-channel of frame N. Instead, it is transmitted in the side-channel of frame N+j, where j is an offset determined by a pre-shared secret or a pseudo-random seed. An attacker who intercepts the communication stream would not be able to determine which key corresponds to which data packet, rendering the intercepted information useless.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its Federal law enforcement agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the embedding of covert side-channels within a primary high-cardinality data stream.

FIG. 2 is a diagram illustrating the method of phase-shifting the side-channel data for enhanced security.

Detailed Description of the Invention

Referring to **FIG. 1**, a high-cardinality parallel data stream (100) is shown, comprising a sequence of data frames (101). Each frame consists of a primary data channel (102) and one or more covert side-channels (103). The primary data, such as the main payload of a message, is transmitted in the primary channel. Sensitive information, such as an encryption key (104), is transmitted in the covert side-channel. While traveling on the same physical medium, the channels are logically separate. An attacker monitoring the primary data stream (102) would not be aware of the existence of the side-channels (103). For a message to be reconstructed, both the primary data and the side-channel data must be received and correctly combined (105).

Referring to **FIG. 2**, the method of phase-shifting is illustrated. A stream of data frames (200) is transmitted. The primary data for frame 1 (D1) is transmitted in the primary channel of frame 1. However, the corresponding encryption key (K1) is not transmitted in the side-channel of frame 1. Instead, its transmission is delayed by an offset, j (201). In this example, with j=2, the key K1 is transmitted in the side-channel of frame 3. Similarly, the key for frame 2 (K2) is transmitted in frame 4, and so on. A receiving system with knowledge of the offset j can correctly re-associate the keys with their corresponding data frames. An attacker without this knowledge would be unable to decrypt the data, even if they could intercept and distinguish both the primary and side-channels. The offset j can be a fixed, pre-shared secret, or it can be a variable determined by a pseudo-random sequence known to both the sender and receiver.

Claims

What is claimed is:

1. A system for secure data communication, comprising:
    - a high-cardinality, parallel data interface configured to transmit a stream of data frames;
    - wherein each data frame comprises a primary data channel and at least one covert side-channel, which are logically separate but share a common physical transmission medium; and
    - an encoding module configured to embed a primary data payload into the primary data channel and to embed sensitive auxiliary data, such as a cryptographic key, into the covert side-channel.
2. The system of claim 1, wherein the parallel data interface is an optical interface.
3. The system of claim 2, wherein the primary data is encoded in the wavelengths of light in the optical interface, and the covert side-channel data is encoded in the spatial arrangement of the light.
4. The system of claim 1, wherein the encoding module is further configured to transmit the auxiliary data for a given data frame in the covert side-channel of a subsequent data frame, delayed by a predetermined offset.

5. The method of claim 4, wherein the offset is determined by a pseudo-random sequence known to the transmitter and a receiver.

6. A method for secure data communication, comprising the steps of:

    - transmitting a primary data payload in a primary channel of a high-cardinality, parallel data stream; and
    - transmitting sensitive auxiliary data corresponding to the primary data payload in a covert side-channel that is logically separate from but physically shares a medium with the primary channel.

7. The method of claim 6, further comprising the step of delaying the transmission of the auxiliary data by a predetermined offset of data frames relative to the transmission of the corresponding primary data payload.

Drawings

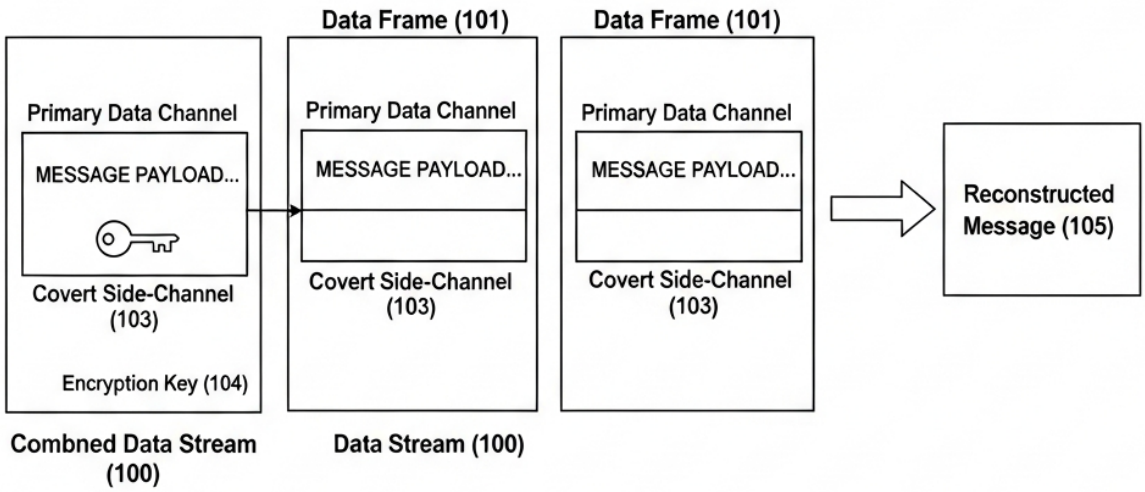


Figure 1: Conceptual diagram of embedded covert side-channels.

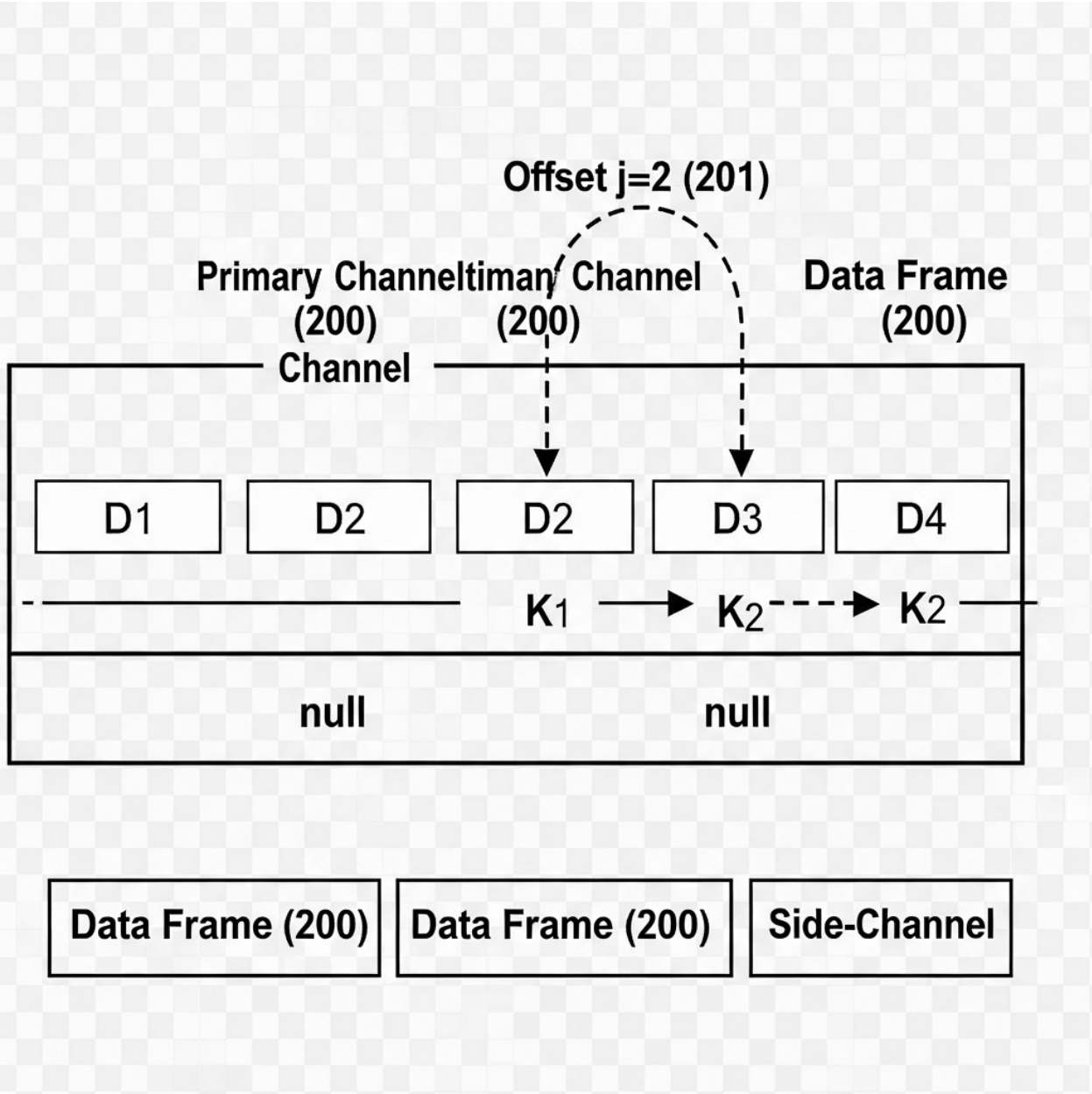


Figure 2: Diagram illustrating phase-shifting of side-channel data.